

This finishing technique is especially useful when working with 3D printers that use multiple materials in their process. Subtractive printing allows users to flesh out the variances in composition and style – especially handy when multiple colour components or parts are printed together. Another aspect of the finishing process is erasing or removing “supports” that are used while printing. In certain designs, it’s necessary to use small components that hold up the orientation or structure of the object during the printing process, after which they’re no longer useful and need to be removed. During the finishing process, these supports are removed either manually or by dissolving them once the object has been printed.

Additives used

We’ve said it before and we’ll say it again: 3D printing is essentially additive printing where materials are added in layers to achieve the final object. These materials that are used by 3D printers are called additives, which are available in a variety of types contingent on the type of printer used and the object designed. Additives are essentially the “ink” with which three-dimensional objects are printed. A significantly large variety of materials – ranging from plastic to chocolate – can be used as additives, if the printer supports them of course.

The most common and affordable additive material is plastic of various properties. The plastic used to make LEGO toys is one of the top used additives and is called acrylonitrile butadiene styrene or ABS. It is cheaply available and easy to work with, in addition to being found in a variety of colours. The other types of plastics being used in 3D printing are as follows:-

1. **Polylactic acid-based plastics (PLA):** Available in a range of grades, from soft to hard. It’s gradually gaining favour as the more preferred additive type in comparison to ABS.
2. **Polyvinyl alcohol (PVA):** It’s an essential additive agent used as a dissolvable material, most commonly used to make supports in 3D objects.
3. **Polycarbonate (PC):** Polycarbonate is a currently experimental additive material being tested with certain types of 3D printers. It’s ejected in liquid state from high temperature ejectors or printer nozzles, similar to inkjet printers.
4. **Soft poly-lactic acid (Soft PLA):** This plastic is the extreme version of PLAs and has the unique property of being very flexible. However, it’s currently still in limited use due to its basic range of colours and production.